Modelling neglect-zero in team semantics

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Slides: https://www.marialoni.org/resources/2BNU26.pdf











Beijing Normal University

Lecture 2, 16 October 2026

NØthing is logical (Nihil)

Outline of the lectures

1. General introduction: Neglect-zero & no-split [Mon, 12 Oct]
2. **Modelling neglect-zero in teams semantics** [Fri, 16 Oct]
3. Experimenting with zero-models [Mon, 19 Oct]

Literature

- Aloni, Maria (2022). "Logic and conversation: The case of free choice". In: *Semantics and Pragmatics* 15.5, pp. 1–60.
- Aloni, Maria, Aleksi Anttila, and Fan Yang (2024). "State-based Modal Logics for Free Choice". In: *Notre Dame Journal of Formal Logic* 65.4, pp. 367–413.
- Søren Knudstorp, Maria Aloni, Alexandru Baltag, Johan van Benthem, Nick Bezhanishvili (2026). *Teams, Truthmakers, Translations*. `https://arxiv.org/abs/2610.15680`

The neglect-zero hypothesis

NEGLECT-ZERO
When interpreting a sentence speakers construct models depicting reality and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*)

Illustrations

(1) Less than three squares are black. **[ESQ]**

a. Verifier: [■, □, ■] $\rightsquigarrow_{nz}$ there are black squares
b. Falsifier: [■, ■, ■]
c. **Zero-models**: [□, □, □]; [■, ■, ■]; . . .

Zero-models in (1-c) verify the sentence by virtue of an empty set of black squares

(2) Maria ate an apple or a banana. **[Possibility]**

a. Verifier: [🍎 | 🍌] ⇐ 'split' $\rightsquigarrow_{nz}$ it might be apple & it might be banana
b. Falsifiers: [🍐]; [ ]
c. **Zero-models**: [🍎]; [🍌]

Zero-models in (2-c) verify the sentence by virtue of an empty witness for one of the disjuncts

The neglect-zero hypothesis

> Neglect-zero
> When interpreting a sentence speakers construct models depicting reality and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*)

NZ hypothesis

Neglect-zero at the origin of many common departures from classical reasoning

- Free choice: possible (A or B) $\Rightarrow$ possible A
- Aristotle's Thesis: if not A then A $\Rightarrow \bot$
- Boethius' Thesis: if A then B & if A then not B $\Rightarrow \bot$
- Existential Import: every A is B $\Rightarrow$ some A is B
- and more

Today

1. Modelling neglect-zero in a team semantics: BSML and variants
   - Free choice
   - Paradoxes of implication
2. Appendix: Comparisons via translation in extensions of MIL

Paradox of Free Choice

Challenge 1: Explosion

- FC inference invalid in classical modal logic but robust in natural language:

  (3) a. You can have cake or ice cream $\rightsquigarrow$ You can have cake and you can have ice cream

  b. FC: $\Diamond(a \vee b) \rightsquigarrow \Diamond a \wedge \Diamond b$

- Adding FC to classical modal logic (ML) implies the equivalence of any two possibility claims [Kamp 1973]

$$\Diamond a \Rightarrow_{\textit{or}\text{-intro+mon}} \Diamond(a \vee b) \Rightarrow_{\text{FC}} \Diamond b$$

Challenge 2: Double Prohibition (DP)

- FC effects systematically disappear in negative contexts [Alonso-Ovalle 2006]

  (4) a. You can't have cake or ice cream $\rightsquigarrow$ You can't have cake and you can't have ice cream

  b. DP: $\neg\Diamond(a \vee b) \rightsquigarrow \neg\Diamond a \wedge \neg\Diamond b$

- The combination of FC & DP leads again to explosion [Goldstein 2019]

FC: $\Diamond(a \vee b) \models \Diamond b \Rightarrow_{\textit{contraposition}} \neg\Diamond b \models \neg\Diamond(a \vee b)$

DP: $\neg\Diamond(a \vee b) \models \neg\Diamond a \Rightarrow_{\textit{transitivity}} \neg\Diamond b \models \neg\Diamond a \Rightarrow_{\textit{contraposition}} \Diamond a \models \Diamond b$

Paradox of Free Choice

Challenge 3: Universal FC

- FC effects can embed under universal quantifiers [Chemla 2009]

(5) a. All of the boys may go to the beach or to the cinema $\rightsquigarrow$ All of the boys may go to the beach and all of the boys may go to the cinema

b. U-FC: $\forall x \Diamond(\alpha \vee \beta) \rightsquigarrow \forall x(\Diamond\alpha \wedge \Diamond\beta)$

Challenge 4: Wide Scope FC

- FC also arise with wide scope disjunctions [Zimmermann 00, Cremers *et al* 17]

(6) a. Detectives may go by bus or they may go by boat
$\rightsquigarrow$ Detectives may go by bus and may go by boat

b. Mr. X might be in Victoria or he might be in Brixton
$\rightsquigarrow$ Mr. X might be in Victoria and might be in Brixton

c. WS-FC: $\Diamond\alpha \vee \Diamond\beta \rightsquigarrow \Diamond\alpha \wedge \Diamond\beta$

	FC	D prohibition	Universal FC	WS-FC
Semantic account	yes	**no**	yes	?
Pragmatic account	yes	yes	**no**	?

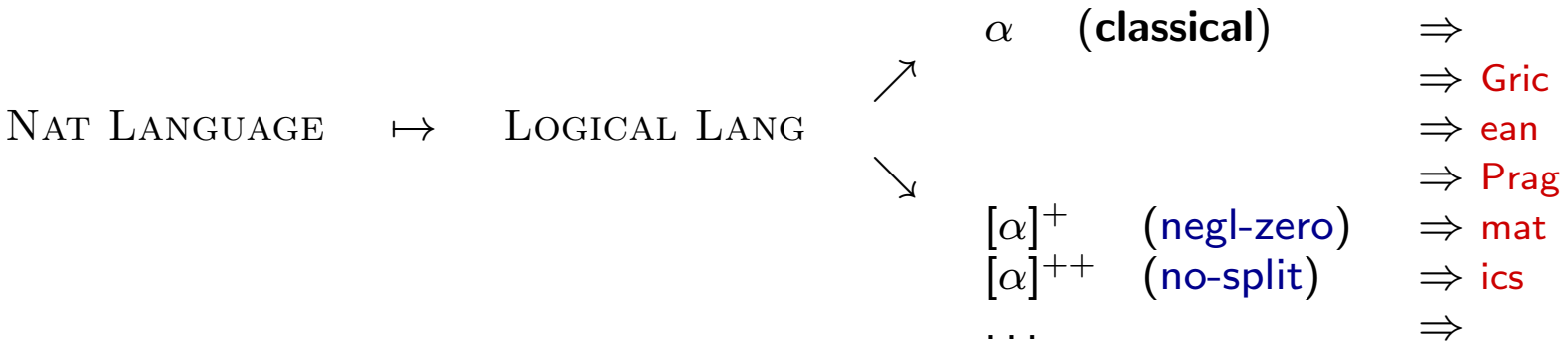
Table: Purely semantic and pragmatic account of FC.

Goal: meet these challenges in cognitive bias approach where FC inference are neglect-zero effects

Modelling biases in team semantics

General methodology

Natural language sentences translated into classical logic formulas interpreted in a team semantics which models both classical and bias-driven interpretations



Back to FC challenge 1

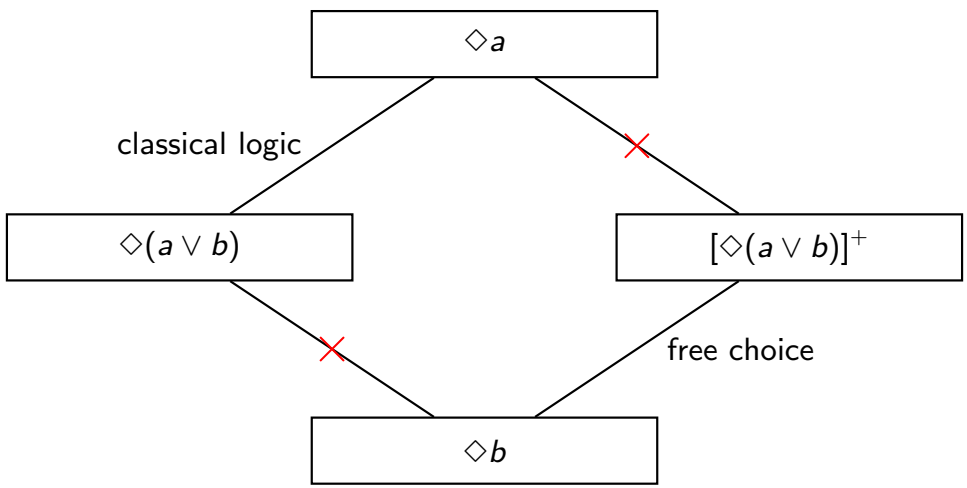


Figure: FC derived only for neglect-zero (NZ) enriched formulas

BSML: teams and bilateralism

- Team semantics: formulas interpreted wrt a set of points of evaluation (a team) rather than single ones [Väänänen 2007; Yang & Väänänen 2017]

Classical vs team-based modal logic

$[M = \langle W, R, V \rangle]$

- Classical modal logic: (truth in worlds)

$$M, w \models \phi, \text{ where } w \in W$$

- Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

Bilateral state-based modal logic (BSML)

- Teams $\mapsto$ information states [Dekker93; Groenendijk$^+$96; Ciardelli$^+$19]
- Assertion & rejection conditions modeled rather than truth

$$M, s \models \phi, \text{ “}\phi \text{ is assertable in } s\text{”}, \quad \text{with } s \subseteq W$$

$$M, s \models\!\!\!| \phi, \text{ “}\phi \text{ is rejectable in } s\text{”}, \quad \text{with } s \subseteq W$$

Modelling neglect-zero

- Crucial feature: **empty team** among possible teams/states: $\emptyset \subseteq W$
- Neglect-zero modelled via non-emptiness atom NE which disallows empty teams as possible verifiers:

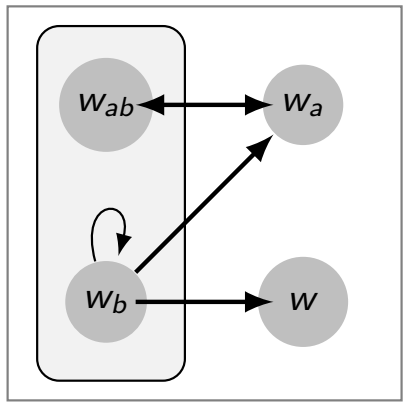
$$M, s \models \text{NE} \text{ iff } s \neq \emptyset$$

- **Bilateralism** crucial to derive cancellation of neglect-zero effects under negation

BSML: Classical Modal Logic + NE

Language

$$\phi := p \mid \neg\phi \mid \phi \vee \phi \mid \phi \wedge \phi \mid \Diamond\phi \mid \text{NE}$$



Bilateral team semantics

Given Kripke model $M = \langle W, R, V\rangle$ & teams/states $s, t, t' \subseteq W$

$M, s \models p$	iff	for all $w \in s : V(w, p) = 1$
$M, s \mathrel{=\!\!|} p$	iff	for all $w \in s : V(w, p) = 0$
$M, s \models \neg\phi$	iff	$M, s \mathrel{=\!\!|} \phi$
$M, s \mathrel{=\!\!|} \neg\phi$	iff	$M, s \models \phi$
$M, s \models \phi \vee \psi$	iff	there are $t, t' : t \cup t' = s$ & $M, t \models \phi$ & $M, t' \models \psi$ $\Leftarrow$
$M, s \mathrel{=\!\!|} \phi \vee \psi$	iff	$M, s \mathrel{=\!\!|} \phi$ & $M, s \mathrel{=\!\!|} \psi$
$M, s \models \phi \wedge \psi$	iff	$M, s \models \phi$ & $M, s \models \psi$
$M, s \mathrel{=\!\!|} \phi \wedge \psi$	iff	there are $t, t' : t \cup t' = s$ & $M, t \mathrel{=\!\!|} \phi$ & $M, t' \mathrel{=\!\!|} \psi$
$M, s \models \Diamond\phi$	iff	for all $w \in s$: there is $t \subseteq R[w] : t \neq \emptyset$ & $M, t \models \phi$
$M, s \mathrel{=\!\!|} \Diamond\phi$	iff	for all $w \in s : M, R[w] \mathrel{=\!\!|} \phi$ [where $R[w] = \{v \in W \mid wRv\}$]
$M, s \models \text{NE}$	iff	$s \neq \emptyset$
$M, s \mathrel{=\!\!|} \text{NE}$	iff	$s = \emptyset$

Illustrations $\{w_{ab}, w_b\} \not\models a$; $\not\models \neg a$, but $\models a \vee \neg a$; $\{w_{ab}, w_b\} \models b$; but $\not\models \Diamond b$; $\models \Diamond a$;
$\emptyset \models a$; $\emptyset \models b$; $\emptyset \mathrel{=\!\!|} a$; $\emptyset \mathrel{=\!\!|} b$; $\emptyset \models \Diamond a$; $\emptyset \models \Diamond b$; $\emptyset \mathrel{=\!\!|} \Diamond a$; $\emptyset \mathrel{=\!\!|} \Diamond b$; $\emptyset \not\models \text{NE}$

Fact (**Empty team property**). Every NE-free (classical) formula α in BSML is supported & anti-supported by the empty team: $M, \emptyset \models \alpha$ & $M, \emptyset \mathrel{=\!\!|} \alpha$

BSML: some results

Proof Theory: BSML [MA, Anttila & Yang, 2024]
`https://doi.org/10.1215/00294527-2024-0027`

Expressive completeness: BSML exp. complete for the class of all convex union-closed modal properties invariant under bounded bisimulation [Anttila & Knudstorp 25]
`https://arxiv.org/abs/2503.21850`

Comparisons via translation into MIL: BSL$_\blacklozenge$ [Knudstorp *et al* 2026]
`https://arxiv.org/abs/2610.15680`

Proof Theory: BSL$^*_\blacklozenge$ [Menorello 2026]
`https://www.illc.uva.nl/Research/Publications/Reports/MoL/`



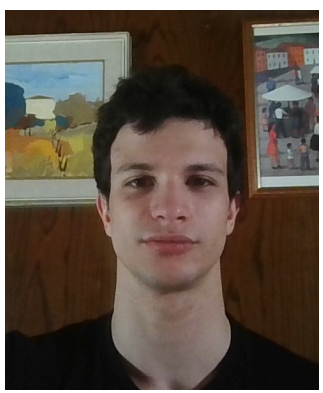
Aleksi Anttila



Søren Knudstorp



Fan Yang



E. Menorello

Neglect-zero effects in BSML: neglect-zero enrichment

Neglect-zero modelled syntactically using NE

Non-emptiness

NE is supported in a state if and only if the state is not empty

$$M, s \models \mathrm{NE} \quad \text{iff} \quad s \neq \emptyset$$
$$M, s \not\models \mathrm{NE} \quad \text{iff} \quad s = \emptyset$$

Neglect-zero enrichment function $[\ ]^+$

For NE-free (classical) α, $[\alpha]^+$ defined as follows:

$$\begin{aligned}
[p]^+ &= p \wedge \mathrm{NE} \\
[\neg\alpha]^+ &= \neg[\alpha]^+ \wedge \mathrm{NE} \\
[\alpha \vee \beta]^+ &= ([\alpha]^+ \vee [\beta]^+) \wedge \mathrm{NE} \\
[\alpha \wedge \beta]^+ &= ([\alpha]^+ \wedge [\beta]^+) \wedge \mathrm{NE} \\
[\Diamond\alpha]^+ &= \Diamond[\alpha]^+ \wedge \mathrm{NE}
\end{aligned}$$

$[\ ]^+$ enriches formulas with the requirement to satisfy NE distributed along each of their subformulas

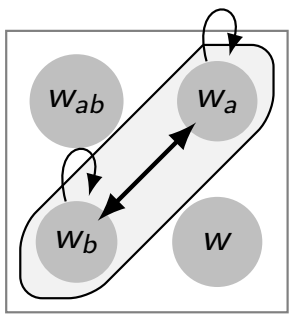
Neglect-zero effects in BSML: disjunction

$M, s \models \phi \vee \psi$ iff there are $t, t' : t \cup t' = s$ & $M, t \models \phi$ & $M, t' \models \psi$

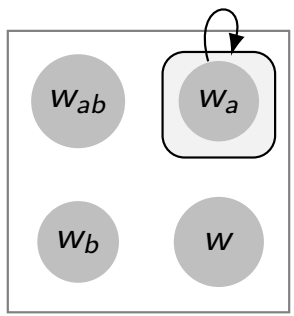
$M, s \models \phi \wedge \psi$ iff $M, s \models \phi$ & $M, s \models \psi$

- s supports a **classical disjunction** $\alpha \vee \beta$ iff s is the union of two (possibly empty) substates, each supporting one of the disjuncts
- s supports an **enriched disjunction** $[\alpha \vee \beta]^+$ iff s is the union of two non-empty substates, each supporting one of the disjuncts

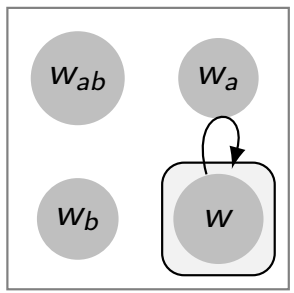
$$[\alpha \vee \beta]^+ = (\alpha \wedge \text{NE}) \vee (\beta \wedge \text{NE}) \wedge \text{NE}$$



No-zero verifier
$\models a \vee b$; $\models [a \vee b]^+$



Zero-model
$\models a \vee b$; $\not\models [a \vee b]^+$



Falsifier
$\not\models a \vee b$; $\not\models [a \vee b]^+$

- An enriched disjunction requires both disjuncts to be live possibilities

[Zimmermann 2000]

(7) M ate an apple or a banana $\rightsquigarrow_{nz}$ It might be apple & it might be banana

$[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$ (where R is state-based)

Neglect-zero effects in BSML: disjunction

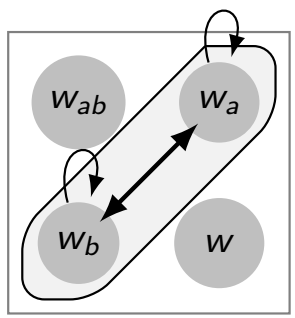
$$M, s \Mid\!= \phi \vee \psi \quad \text{iff} \quad M, s \Mid\!= \phi \ \&\ M, s \Mid\!= \psi$$
$$M, s \Mid\!= \phi \wedge \psi \quad \text{iff} \quad \text{there are } t, t' : t \cup t' = s \ \&\ M, t \Mid\!= \phi \ \&\ M, t' \Mid\!= \psi$$

- s anti-supports a **classical disjunction** $\alpha \vee \beta$ iff $s \Mid\!= \alpha$ and $s \Mid\!= \beta$
- s anti-supports an **enriched disjunction** $[\alpha \vee \beta]^+$ iff $s \Mid\!= \alpha$ and $s \Mid\!= \beta$

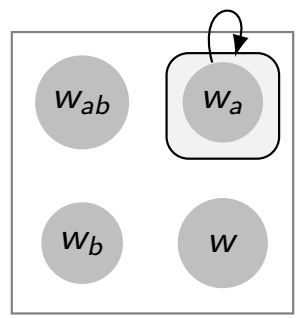
$$[\alpha \vee \beta]^+ = (\alpha \wedge \text{NE}) \vee (\beta \wedge \text{NE}) \wedge \text{NE}$$

Crucial fact: $s \Mid\!= (\phi \wedge \text{NE}) \Leftrightarrow s \Mid\!= \phi$

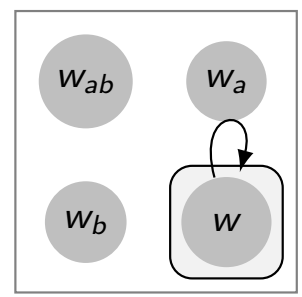
Proof. $s \Mid\!= \phi \wedge \text{NE} \Leftrightarrow s = t \cup t' : t \Mid\!= \phi \ \&\ t' \Mid\!= \text{NE}$. But only $\emptyset \Mid\!= \text{NE}$. Therefore s must be t.



No-zero verifier
$\models a \vee b$; $\models [a \vee b]^+$
$\not\Mid\!= a \vee b$; $\not\Mid\!= [a \vee b]^+$



Zero-model
$\models a \vee b$; $\not\models [a \vee b]^+$
$\not\Mid\!= a \vee b$; $\not\Mid\!= [a \vee b]^+$



Falsifier
$\not\models a \vee b$; $\not\models [a \vee b]^+$
$\Mid\!= a \vee b$; $\Mid\!= [a \vee b]^+$

- We derive strong 'neither' readings for enriched disjunction under negation:

(8) M didn't an apple or a banana ⇒ M didn't eat either
$[\neg(\alpha \vee \beta)]^+ \models \neg\alpha \wedge \neg\beta$

Neglect-zero effects in BSML: predictions

Main result

In BSML $[\,]^+$-enrichment has non-trivial effect only when applied to *positive* disjunctions [MA 2022]

- $\mapsto$ we derive possibility, FC and related effects (for enriched formulas);
- $\mapsto$ $[\,]^+$-enrichment vacuous under single negation.

Neglect-zero effects in BSML: predictions

After enrichments

- We derive possibility and FC inferences for enriched formulas:
  - Possibility: $[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$ (if R is state-based)
  - Narrow scope FC: $[\Diamond(\alpha \vee \beta)]^+ \models \Diamond\alpha \wedge \Diamond\beta$
  - Universal FC: $[\forall x \Diamond(\alpha \vee \beta)]^+ \models \forall x(\Diamond\alpha \wedge \Diamond\beta)$ (in qBSML)
  - Wide scope FC: $[\Diamond\alpha \vee \Diamond\beta]^+ \models \Diamond\alpha \wedge \Diamond\beta$ (if R is indisputable)
- while no undesirable side effects obtain with other configurations:
  - Double prohibition: $[\neg\Diamond(\alpha \vee \beta)]^+ \models \neg\Diamond\alpha \wedge \neg\Diamond\beta$

    [Chall. 2: failure of contraposition: $[\Diamond(a \vee b)]^+ \models \Diamond b \not\Rightarrow \neg\Diamond b \models \neg[\Diamond(a \vee b)]^+$]

Before enrichments

- The NE-free fragment of BSML is equivalent to classical modal logic (ML):

$$\alpha \models_{BSML} \beta \text{ iff } \alpha \models_{ML} \beta \qquad [\text{if } \alpha, \beta \text{ are NE-free}]$$

- But we can capture infelicity of epistemic contradictions (Yalcin, 2007) by putting team-based constraints on accessibility relation:

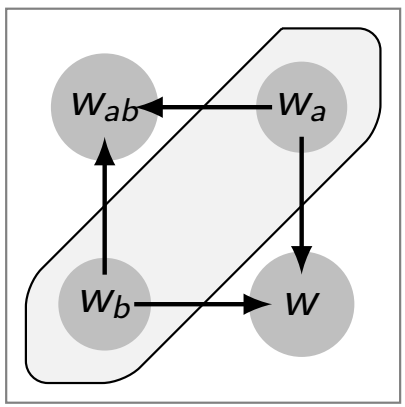
(9) #It might be raining and it is not raining.

1. Epistemic contradiction: $\Diamond_e \alpha \wedge \neg\alpha \models \bot$ (if R is state-based)
2. Non-factivity: $\Diamond_e \alpha \not\models \alpha$

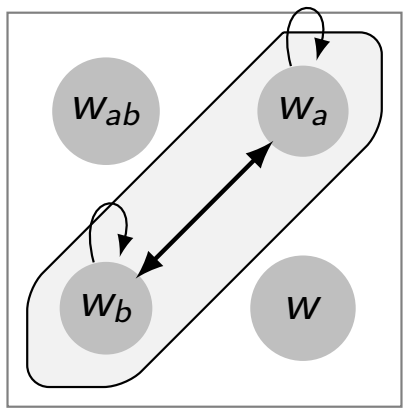
Neglect-zero effects in BSML: last ingredient

Team-sensitive constraints on accessibility relation

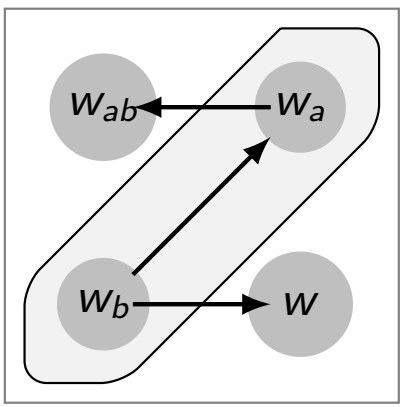
- R is indisputable in (M, s) iff $\forall w, v \in s : R[w] = R[v]$
  $\mapsto$ all worlds in s access exactly the same set of worlds
- R is state-based in (M, s) iff $\forall w \in s : R[w] = s$
  $\mapsto$ all and only worlds in s are accessible within s



(a) indisputable



(b) state-base (and so also indisputable)



(c) neither

Deontics vs epistemics

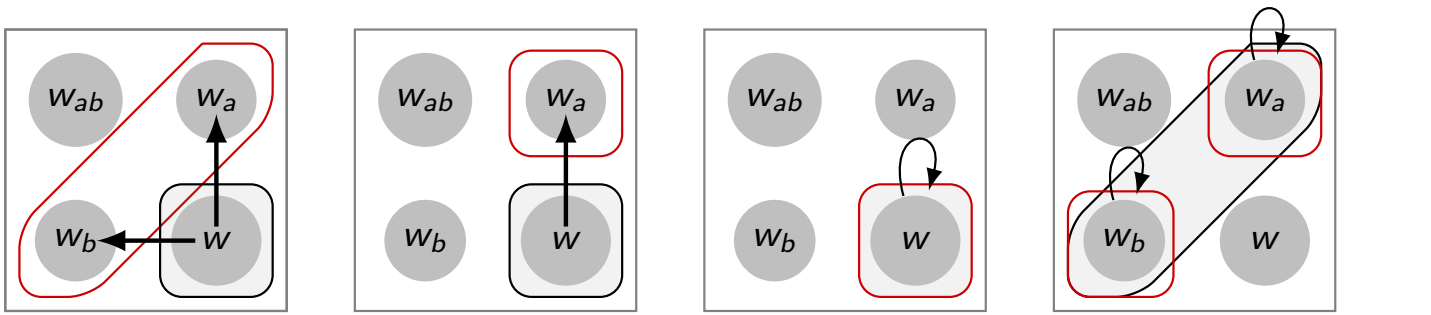
- Difference deontic vs epistemic modals captured by different properties of accessibility relation:
  - Epistemics: R is state-based ($\Rightarrow$ epistemic contradictions)
  - Deontics: R is possibly indisputable (e.g. in performative uses)

BSML: illustration free choice facts

- A team s supports $\Diamond\phi$ iff for every w in s there is a non-empty subset of the set of worlds accessible from w which supports ϕ:

$$M, s \models \Diamond\phi \text{ iff for all } w \in s: \text{ there is } t \subseteq R[w] : t \neq \emptyset \ \&\ M, t \models \phi$$

- **Narrow and wide scope** FC
  - Narrow scope FC: $[\Diamond(\alpha \vee \beta)]^+ \models \Diamond\alpha \wedge \Diamond\beta$
  - Wide-scope FC: $[\Diamond\alpha \vee \Diamond\beta]^+ \models \Diamond\alpha \wedge \Diamond\beta$ [if R is indisputable]



No-zero verifier
$\models \Diamond(a \vee b)$
$\models [\Diamond(a \vee b)]^+$
$\models \Diamond a \vee \Diamond b$
$\models [\Diamond a \vee \Diamond b]^+$
$\models \Diamond a; \models \Diamond b$

Zero model
$\models \Diamond(a \vee b)$
$\not\models [\Diamond(a \vee b)]^+$
$\models \Diamond a \vee \Diamond b$
$\not\models [\Diamond a \vee \Diamond b]^+$
$\models \Diamond a; \not\models \Diamond b$

Falsifier
$\not\models \Diamond(a \vee b)$
$\not\models [\Diamond(a \vee b)]^+$
$\not\models \Diamond a \vee \Diamond b$
$\not\models [\Diamond a \vee \Diamond b]^+$
$\not\models \Diamond a; \not\models \Diamond b$

($*$) not indisputable R
$\models \Diamond(a \vee b)$
$\not\models [\Diamond(a \vee b)]^+$
$\models \Diamond a \vee \Diamond b$
$\models [\Diamond a \vee \Diamond b]^+ \Leftarrow$
$\not\models \Diamond a; \not\models \Diamond b$

Figure: Models for $\Diamond(a \vee b)$ and $\Diamond a \vee \Diamond b$.

- **Failure of wide scope** FC when R is not indisputable:
  - ($*$) zero model for $\Diamond(a \vee b)$, but no-zero verifier for $\Diamond a \vee \Diamond b$
  - ($*$) $\models [\Diamond a \vee \Diamond b]^+$, but ($*$) $\not\models \Diamond a$; ($*$) $\not\models \Diamond b$ (NB: R is not indisputable)

BSML predictions: epistemic and deontic FC

- Narrow scope FC: $[\Diamond(\alpha \vee \beta)]^+ \models \Diamond\alpha \wedge \Diamond\beta$
- Wide-scope FC: $[\Diamond\alpha \vee \Diamond\beta]^+ \models \Diamond\alpha \wedge \Diamond\beta$ [if R is indisputable]

Epistemic modals

- R is **state-based**, therefore always indisputable:

(10) He might be in either London or Paris. [+fc, narrow]
(11) He might be in London or he might be in Paris. [+fc, wide]

$\Rightarrow$ narrow and wide scope FC always predicted for enriched epistemics

Deontic modals

- R **sometimes indisputable**, e.g. in performative uses

(12) You may eat either the cake or the ice-cream. [+fc, narrow]
(13) Either you may eat the cake or you may eat the ice-cream. [-fc, wide]

$\Rightarrow$ narrow scope FC always predicted for enriched deontics

$\Rightarrow$ wide scope FC only if speaker is informed about what is permitted/obligatory [Cremers et al 2017]

Further consequence: all cases of (overt) FC cancellations involve a wide scope configuration in a context where indisputability is not warranted

BSML predictions: overt FC cancellations

- Examples of overt FC cancellations:

  (14) You may eat the cake or the ice-cream, I don't know which $\not\rightsquigarrow$ You may eat the cake

  (15) You may eat the cake or the ice-cream, it depends on what John has taken $\not\rightsquigarrow$ You may eat the cake [Kaufmann 2016]

- Sluicing in (14) and inquisitive *it* in (15) arguably triggers wide scope disjunction in their antecedent [Fusco 2019, Pinton & MA 2022]

  (16) You may eat the cake or the ice-cream, I don't know which (you may eat). [wide, –fc]

  (17) You may eat the cake or the ice-cream, it (= what you may eat) depends on what John has taken. [wide, –fc]

- Sketch of analysis (in BSML^{F} + inquisitive disjunction):

  (a) which/what you may eat $\mapsto \Diamond\alpha \mathbin{/\!\!/} \Diamond\beta \equiv \boxed{\begin{array}{c}\Diamond\alpha\\ \hline \Diamond\beta\end{array}}$

  (b) $\Diamond\alpha \vee \Diamond\beta \equiv \mathrm{F}\boxed{\begin{array}{c}\Diamond\alpha\\ \hline \Diamond\beta\end{array}} \not\equiv \Diamond(\alpha \vee \beta)$

BSML predictions: negation

- **Negation** facts follow from adopted bilateralism (we validate De Morgan laws and $\neg\neg$-elimination):
  - Adding NE vacuous under single negation:

$$\neg(\alpha \wedge \text{NE}) \equiv \neg\alpha \vee \neg\text{NE} \equiv \neg\alpha \vee \bot \equiv \neg\alpha$$

  - Adding NE non-vacuous under double negation:

$$\neg\neg(\alpha \wedge \text{NE}) \equiv \alpha \wedge \text{NE} \not\equiv \neg\neg\alpha$$

⇒ Failure of replacement under negation:

$$\phi \equiv \psi \not\Rightarrow \neg\phi \equiv \neg\psi$$

⇒ Empirically correct predictions:
  - FC effects systematically disappear under **single negation** [Alonso-Ovalle 2006]
  - But speakers draw FC inferences under **double negation** [Romoli & Santorio 2019, Gotzner et al. 2020]

(18) Double Negation FC [Gotzner *et al.* 2020]

a. Exactly one girl cannot take Spanish or Calculus ⇝ One girl can take neither of the two and each of the others can choose between them.

b. $\exists x(\neg\Diamond(\alpha(x) \vee \beta(x)) \wedge \forall y(y \neq x \rightarrow \neg\neg\Diamond(\alpha(y) \vee \beta(y)))) \rightsquigarrow$
$\exists x(\neg\Diamond\alpha(x) \wedge \neg\Diamond\beta(x) \wedge \forall y(y \neq x \rightarrow (\Diamond\alpha(y) \wedge \Diamond\beta(y))))$

⇒ Without replacement failure impossible to capture these data

Neglect-zero effects in BSML: further predictions

D vs UN inferences

- **Possibility** and **Distributivity (D)** inferences derived as neglect-zero effects:
  - $[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$ (possibility)
  - $[\Box(\alpha \vee \beta)]^+ \models \Diamond\alpha \wedge \Diamond\beta$ ($\Box$-FC)
  - $[\forall x(\alpha \vee \beta)]^+ \models \exists x\alpha \wedge \exists x\beta$ (qBSML)
- **Uncertainty** and **Negative Universal (NU)** inferences not derived:
  - $[\alpha \vee \beta]^+ \not\models \neg\Box_e \alpha \wedge \neg\Box_e \beta$ (uncertainty)
  - $[\Box(\alpha \vee \beta)]^+ \not\models \neg\Box\alpha \wedge \neg\Box\beta$
  - $[\forall x(\alpha \vee \beta)]^+ \not\models \neg\forall x\alpha \wedge \neg\forall x\beta$ (qBSML)

[Crnic, Chemla & Fox, 2015; Ramotowska et al, 2022; Degano et al, 2025]

Negative FC

- **Negative** FC not predicted in BSML$^+$: [Marty et al 2022]
  - $[\neg(\alpha \wedge \beta)]^+ \not\models \Diamond_e\neg\alpha \wedge \Diamond_e\neg\beta$
  - $[\neg\Box(\alpha \wedge \beta)]^+ \not\models \Diamond\neg\alpha \wedge \Diamond\neg\beta$ [$\neq$ BSML* ($\emptyset$ not among possible teams)]
- **Experimental research**: negative FC inferences exist but appear to be less available than positive FC
- (BSML* needed to capture children's 'neither' readings of negated conjunction)

Two neglect-zero enrichment functions

For NE-free α,

- $[\alpha]^+$ defined as follows:

$$\begin{aligned}
[p]^+ &= p \wedge \mathrm{NE} \\
[\neg\alpha]^+ &= \neg[\alpha]^+ \wedge \mathrm{NE} \\
[\alpha \vee \beta]^+ &= ([\alpha]^+ \vee [\beta]^+) \wedge \mathrm{NE} \\
[\alpha \wedge \beta]^+ &= ([\alpha]^+ \wedge [\beta]^+) \wedge \mathrm{NE} \\
[\Diamond\alpha]^+ &= \Diamond[\alpha]^+ \wedge \mathrm{NE}
\end{aligned}$$

- $[\alpha]^*$ defined as follows: [Klochowicz 2026]

$$\begin{aligned}
[p]^* &= (p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^* &= \neg([\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^* &= (([\alpha]^* \vee [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^* &= (([\alpha]^* \wedge [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^* &= (\Diamond[\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

[*-implementation $\equiv$ to ruling out $\emptyset$ from the set of possible teams]

- Different behaviour under negation:

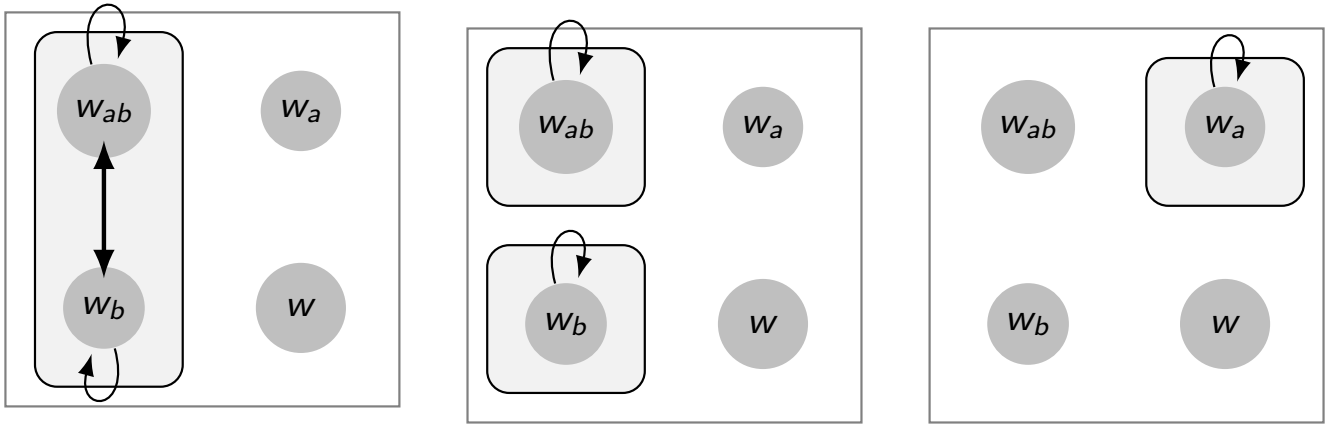
$\neg[p]^+ = \neg(p \wedge \mathrm{NE}) = \neg p \vee \neg\mathrm{NE} = \neg p \vee \bot = \neg p$

$\neg[p]^* = \neg((p \wedge \mathrm{NE}) \vee \neg\mathrm{NE}) = \neg(p \wedge \mathrm{NE}) \wedge \neg\neg\mathrm{NE} = (\neg p \vee \neg\mathrm{NE}) \wedge \mathrm{NE} = \neg p \wedge \mathrm{NE}$

BSML$^\rightarrow$: adding implication

- $\phi \rightarrow \psi = \neg\phi \vee (\phi \wedge \psi)$ (rather than $= \neg\phi \vee \psi$)
  - $M, s \models \phi \rightarrow \psi$ iff there are $t, t' : s = t \cup t'$ & $t \models \neg\phi$ & $t' \models \phi \wedge \psi$
  - $M, s \not\models \phi \rightarrow \psi$ iff $M, s \not\models \neg\phi$ & $M, s \not\models \phi \wedge \psi$
- NZ enrichment via $[\ ]^+$ and not $[\ ]^*$: [What's wrong with $[\ ]^*$?]

$$[\alpha \rightarrow \beta]^+ := (\neg[\alpha]^+ \wedge \text{NE}) \vee (([\alpha]^+ \wedge [\beta]^+) \wedge \text{NE})$$



No-zero verifier
$\models a \rightarrow b$; $\models [a \rightarrow b]^+$

Zero-models
$\models a \rightarrow b$; $\not\models [a \rightarrow b]^+$

Falsifier
$\not\models a \rightarrow b$; $\not\models [a \rightarrow b]^+$

- An enriched implication requires the antecedent and its negation to be live possibilities:

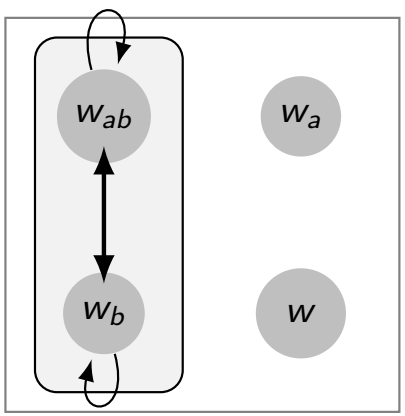
(19) a. If it is raining, M will go home $\rightsquigarrow_{nz}$ it might be raining; it might not be raining

b. $[\alpha \rightarrow \beta]^+ \models \Diamond_e \alpha; \Diamond_e \neg\alpha$ (where R is state-based)

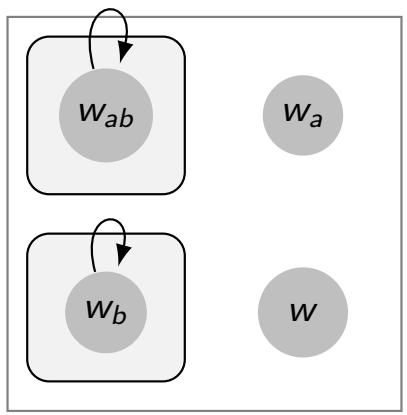
BSML$^\rightarrow$: adding implication

- $\phi \rightarrow \psi = \neg\phi \vee (\phi \wedge \psi)$
  - $M, s \models \phi \rightarrow \psi$ iff there are $t, t' : s = t \cup t'$ & $t \models \neg\phi$ & $t' \models \phi \wedge \psi$
  - $M, s \not\models \phi \rightarrow \psi$ iff $M, s \not\models \neg\phi$ & $M, s \not\models \phi \wedge \psi$
- NZ enrichment via $[\ ]^+$ and not $[\ ]^*$:

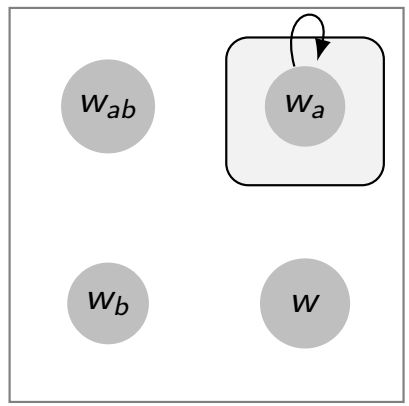
$$[\alpha \rightarrow \beta]^+ := (\neg[\phi]^+ \wedge \text{NE}) \vee (([\phi]^+ \wedge [\psi]^+) \wedge \text{NE})$$



No-zero verifier
$\models a \rightarrow b$; $\models [a \rightarrow b]^+$



Zero-models
$\models a \rightarrow b$; $\not\models [a \rightarrow b]^+$



Falsifier
$\not\models a \rightarrow b$; $\not\models [a \rightarrow b]^+$

- An enriched implication does not follow from its corresponding conjunction (cf. no-link implications):

(20) a. $2 + 2 = 4$ and Paris is the capital of France $\not\rightsquigarrow$ If $2 + 2 = 4$ then Paris is the capital of France

b. $[\alpha \wedge \beta]^+ \not\models [\alpha \rightarrow \beta]^+$

Neglect-zero effects in BSML$^\rightarrow$: other results

After enrichment

- We derive Aristotelean principles for pragmatically enriched formulas:
  - Aristotle's theses: from something we cannot derive its negation

$$[\alpha \rightarrow \neg\alpha]^+ \models \bot \;\&\; [\neg\alpha \rightarrow \alpha]^+ \models \bot$$

  - Abelard/Boethius: from something we cannot derive contradictory conclusions

$$[(\alpha \rightarrow \beta) \wedge (\alpha \rightarrow \neg\beta)]^+ \models \bot$$

  - Existential import: from *every A is B* to *some A is B* [in qBSML]

$$[\forall x(Ax \rightarrow Bx)]^+ \models \exists x(Ax \wedge Bx) \qquad \text{(assuming max info)}$$

Before enrichment

- The NE-free fragment of BSML$^\rightarrow$ is equivalent to classical modal logic:

$$\alpha \models_{BSML^\rightarrow} \beta \text{ iff } \alpha \models_{ML} \beta \qquad [\text{if } \alpha, \beta \text{ are NE-free}]$$

Emerging picture

- Admit zero $\mapsto$ classical logic (literal meanings);
- Disallow zero $\mapsto$ traditional logic (enriched meanings).

The point about zero is that we do not need to use it in the operations of daily life. No one goes out to buy zero fish. It is in a way the most civilized of all the cardinals, and its use is only forced on us by the needs of cultivated modes of thought. (AN Whitehead, quoted by Nieder 2016)

Conclusions

- FC, connexive principles: a mismatch between logic and language
- Grice's insight:
  - stronger meanings can be derived paying more "attention to the nature and importance to the conditions governing conversation"
- Nihil hypothesis: some strengthenings due to cognitive tendencies rather than Gricean reasoning
  - FC, possibility and connexive principles as neglect-zero effects

    Literal meanings (classical fragment) + cognitive factor (NE) $\Rightarrow$ FC, possibility, connexive principles, etc

- Implementation in team-based modal logics
- Appendix: comparison via translations into (extensions of) MIL

THANK YOU![1]

[1]This work is supported by NWO OC project *Nothing is Logical* (grant no 406.21.CTW.023).

Appendix

BSML & related systems: info states (= teams) vs possible worlds

- Failure of bivalence in BSML

$$M, s \not\models p \ \&\ M, s \not\models \neg p, \text{ for some info state } s$$

- Info states: less determinate than possible worlds
  - just like truthmakers, situations, possibilities, HYPE states, ...
- Technically:
  - Truthmakers/possibilities/HYPE states: points in a partially ordered set
  - Info states: sets of possible worlds, also elements of a partially ordered set, the Boolean lattice $Pow(W)$
- Thus systems using these structures are closely connected, although might diverge in motivation:
  - HYPE, truthmaker & possibility semantics: description of ontological structures in the world
  - BSML & inquisitive semantics: explaining patterns in inferential & communicative human activities
- NEXT:
  - Comparison via translations in Modal Information Logic [vBenthem19]

BSML & related systems: comparisons via translation

- Modal Information Logic (MIL) (van Benthem, 1989, 2019):[2] common ground where related systems can be interpreted and their connections and differences can be explored
- Goal: translations into (extensions of) MIL of the following systems:
  - Possibility semantics (Humberstone, Holliday)
  - Truthmaker semantics (Fine)
  - BSML
  - Inquisitive semantics (Ciardelli, Groenendijk & Roelofsen)
  - HYPE (Leitgeb)

  (cf. Gödel's (1933) translation of intuitionistic logic into modal logic)
- Here focus on propositional fragments:
  - disjunction
  - negation
- New result: translation of BSL (propositional fragment of BSML) in an extension of MIL

Søren Knudstorp, Maria Aloni, Alexandru Baltag, Johan van Benthem, Nick Bezhanishvili (2026). *Teams, Truthmakers, Translations*. `https://arxiv.org/abs/2610.15680`

[2] van Benthem, Implicit and Explicit Stances in Logic, *Journal of Philosophical Logic* (2019)

Modal Information Logic (MIL)

Language

$$\phi \quad ::= \quad p \mid \neg\phi \mid \phi \wedge \phi \mid \phi \vee \phi \mid \langle sup \rangle \phi\psi$$

where $p \in \mathrm{P}_\pm := \{p_+, p_- \mid p \in \mathrm{P}\}$.

Models and interpretation

Formulas are interpreted on triples $M = (X, \leq, V)$ where $\leq$ is a partial order

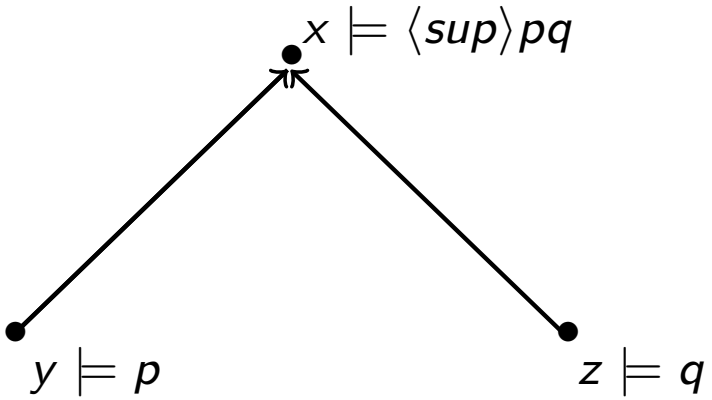
$$\begin{array}{rll}
\mathcal{M}, x \models p & \text{iff} & x \in V(p) \\
\mathcal{M}, x \models \neg\phi & \text{iff} & \mathcal{M}, x \not\models \phi \\
\mathcal{M}, x \models \phi \wedge \psi & \text{iff} & \mathcal{M}, x \models \phi \text{ and } \mathcal{M}, x \models \psi \\
\mathcal{M}, x \models \phi \vee \psi & \text{iff} & \mathcal{M}, x \models \phi \text{ or } \mathcal{M}, x \models \psi \\
\mathcal{M}, x \models \langle sup \rangle \phi\psi & \text{iff} & \text{there are } y, z : x = sup_\leq(y, z) \ \&\ \mathcal{M}, y \models \phi \ \&\ \mathcal{M}, z \models \psi
\end{array}$$

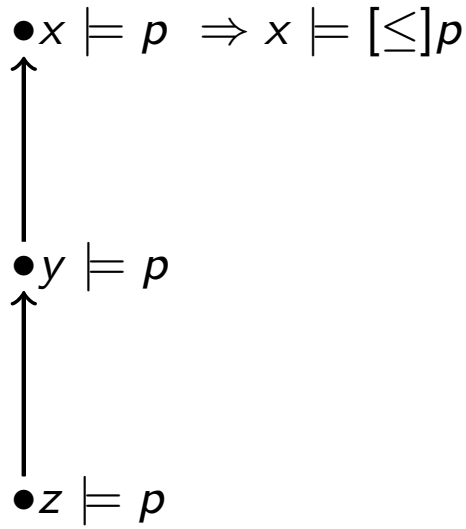
$[\leq]\phi = \neg\langle sup \rangle(\neg\varphi)\top$

$$\mathcal{M}, x \models [\leq]\phi \quad \text{iff} \quad \text{for all } y : y \leq x \ \Rightarrow\ \mathcal{M}, y \models \phi$$

Modal Information Logic (MIL)

Examples





Translations into Modal Information Logic

- Possibility semantics (Humberstone, Holliday)[3]

$$\begin{aligned} tr(p) &= [\leq]\langle\leq\rangle p \\ tr(\neg\phi) &= [\leq]\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= [\leq]\langle\leq\rangle(tr(\phi) \vee tr(\psi)) \\ &\ldots \end{aligned}$$

- Truthmaker semantics (Fine)[4] (MIL on (distributive) semilattices)

$$\begin{aligned} (p)^+ &= p_+ \\ (p)^- &= p_- \\ (\neg\phi)^+ &= (\phi)^- \\ (\neg\phi)^- &= (\phi)^+ \\ (\phi \wedge \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\ (\phi \wedge \psi)^- &= (\phi)^- \vee (\psi)^- \\ (\phi \vee \psi)^+ &= (\phi)^+ \vee (\psi)^+ \\ (\phi \vee \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^- \end{aligned}$$

[3] van Benthem, Bezhanishvili, Holliday, A bimodal perspective on possibility semantics. *Journal of Logic and Computation* (2017)

[4] van Benthem, Implicit and Explicit Stances in Logic. *Journal of Philosophical Logic* (2019)

Translations into (extended) Modal Information Logic

- Truthmaker semantics (Fine) (MIL on (distributive) semilattices)

$$
\begin{aligned}
(p)^+ &= p_+ \\
(p)^- &= p_- \\
(\neg\phi)^+ &= (\phi)^- \\
(\neg\phi)^- &= (\phi)^+ \\
(\phi \wedge \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\
(\phi \wedge \psi)^- &= (\phi)^- \vee (\psi)^- \\
(\phi \vee \psi)^+ &= (\phi)^+ \vee (\psi)^+ \\
(\phi \vee \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^-
\end{aligned}
$$

- Classical fragment BSL (MIL* on distributive semilattices)

$$
\begin{aligned}
(p)^+ &= [\leq]\langle \mathrm{sup}^*\rangle p_+ \\
(p)^- &= [\leq]\langle \mathrm{sup}^*\rangle p_- \\
(\neg\phi)^+ &= (\phi)^- \\
(\neg\phi)^- &= (\phi)^+ \\
(\phi \wedge \psi)^+ &= (\phi)^+ \wedge (\psi)^+ \\
(\phi \wedge \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^- \\
(\phi \vee \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\
(\phi \vee \psi)^- &= (\phi)^- \wedge (\psi)^-
\end{aligned}
$$

$\mathcal{M}, x \models \langle \mathrm{sup}^*\rangle\phi$ iff there exist $x_1, \ldots, x_n$ s.t. each $x_i \models \varphi$ & $x = sup_{\leq}\{x_1 \cdots x_n\}$

Translations into (extended) Modal Information Logic

- Inquisitive semantics (Groenendijk, Roelofsen & Ciardelli)

$$\begin{aligned} tr(p) &= [\leq]\langle \mathrm{sup}^* \rangle p \\ tr(\neg\phi) &= [\leq]\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= tr(\phi) \vee tr(\psi) \\ tr(\phi \rightarrow \psi) &= [\leq](tr(\phi) \rightarrow tr(\psi)) \end{aligned}$$

- HYPE (Leitgeb)[5] (bimodal perspective needed for negation)

$$\begin{aligned} tr(p) &= [\leq]p \\ tr(\neg\phi) &= \blacksquare\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= tr(\phi) \vee tr(\psi) \\ tr(\phi \rightarrow \psi) &= [\leq](tr(\phi) \rightarrow tr(\psi)) \end{aligned}$$

$\mathcal{M}, x \models \blacksquare\phi$ iff for all y: $xCy \Rightarrow \mathcal{M}, y \models \varphi$

Given a compatibility model $\mathcal{M} = (X, C, \leq, V)$

[5] Odintsov, S., Wansing, H. Routley Star and Hyperintensionality. J Philos Logic (2021)

Disjunction and Negation

- Three notions of disjunction expressible in (extended) MIL:
  - Boolean disjunction: $\phi \vee \psi$
    [classical logic, intuitionistic logic, inquisitive logic]
  - Lifted/tensor/split disjunction: $\langle sup \rangle \phi\psi$
    [BSML, dependence logic, team semantics, operational semantics for Positive R]
  - Cofinal disjunction: $[co](\phi \vee \psi)$ (where $[co]\phi =: [\leq]\langle\leq\rangle\phi$)
    [possibility semantics, dynamic semantics]
- Four notions of negation:
  - Boolean negation: $\neg\phi$
    [classical logic, ...]
  - Bilateral negation: $(\neg\phi)^+ = (\phi)^- \ \& \ (\neg\phi)^- = (\phi)^+$
    [truthmaker semantics, BSML, ...]
  - Intuitionistic-like negation: $[\leq]\neg\phi$
    [possibility semantics, inquisitive semantics, intuitionistic logic]
  - Modal negation: $\blacksquare\neg\phi$
    [HYPE, ...]
- Some combinations:
  - Boolean disjunction + boolean negation $\mapsto$ classical logic
  - Boolean notions in other combinations can generate non-classicality:
    - Boolean disjunction + intuitionistic negation $\mapsto$ intuitionistic/inquisitive logic[6]
  - Classicality also generated by non-boolean combinations:
    - Split disjunction + bilateral negation (classical fragm. BSPL)
    - Cofinal disjunction and intuitionistic negation (possibility semantics)

[6] Inquisitive & intuitionistic logic: same connectives but different translations for the atoms.

Full translation for BSPL

$$
\begin{array}{rcl@{\qquad}rcl}
p^+ & := & [\leq]\langle \mathsf{sup}^* \rangle p_+ & p^- & := & [\leq]\langle \mathsf{sup}^* \rangle p_- \\
ne^+ & := & \bigwedge_{p \in \mathrm{P}} \neg(p^+ \wedge p^-) & ne^- & := & \bigwedge_{p \in \mathrm{P}} (p^+ \wedge p^-) \\
\bot^+ & := & ne^- & \bot^- & := & \top \\
(\neg\varphi)^+ & := & \varphi^- & (\neg\varphi)^- & := & \varphi^+ \\
(\varphi \vee \psi)^+ & := & \langle \mathsf{sup} \rangle \varphi^+ \psi^+ & (\varphi \vee \psi)^- & := & \varphi^- \wedge \psi^- \\
(\varphi \wedge \psi)^+ & := & \varphi^+ \wedge \psi^+ & (\varphi \wedge \psi)^- & := & \langle \mathsf{sup} \rangle \varphi^- \psi^-
\end{array}
$$

Theorem (Knudstorp 2026)

$$\varphi \vDash_{BSPL} \psi \qquad \text{iff} \qquad \Gamma, \varphi^+ \vDash_{MIL^*} \psi^+$$

where

$$\Gamma := \left\{ [\leq](ne^+ \vee ne^-), \bigwedge_{p \in \mathrm{P}} \langle \mathsf{sup} \rangle p^+ p^- \right\}$$

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